At the core of their success was something deceptively simple: algorithms built with control flow structures—the same programming constructs we'll explore in this chapter.

Their systems could make millions of lightning-fast decisions, following precisely defined rules without the emotional biases that plague human traders.

Control flow is the programming equivalent of decision-making and repetition—the ability to tell a computer "if this happens, do that" or "repeat this process until a condition is met." It's what transforms code from a linear sequence of instructions into an intelligent system that can respond to changing conditions.

As Howard Marks, co-founder of Oaktree Capital Management with $120 billion under management, once remarked: "The most important thing is to be in control of your emotions." In algorithmic trading, control flow structures are how we codify this wisdom, creating trading systems that follow disciplined rules rather than emotional impulses.

In this chapter, we'll explore three fundamental aspects of control flow:

1. **Conditional statements**: How algorithms make decisions based on market conditions
2. **Loops**: How systems efficiently repeat processes, from data analysis to order execution
3. **Indentation and code readability**: How proper formatting ensures your trading logic works as intended

By mastering these concepts, you'll take a crucial step toward creating trading algorithms that can respond intelligently to market conditions—potentially even achieving your own streak of profitable trading days.